

1) Intro and Recap

I will base these notes on both the lecture notes and the book DSP first. [this](#)

1.1) Complex Numbers

Here is a really quick recap of useful formulas

Magnitude:	$ z = \sqrt{a^2 + b^2}$
Phase:	$\arg z = \arctan \frac{b}{a}$
Complex Conjugate:	$z^* = a - jb$
Euler's Identity:	$e^{jw} = \cos(w) + j\sin(w)$
Polar Representation:	$\cos(w) = \frac{e^{jw} + e^{-jw}}{2}$
	$\sin(w) = \frac{e^{jw} - e^{-jw}}{2j}$

Moreover, let $z_i = a_i + jb_i$, then

$$\begin{aligned}z_1 + z_2 &= (a_1 + a_2) + j(b_1 + b_2) \\z_1 z_2 &= (a_1 a_2 - b_1 b_2) + j(a_1 b_2 + a_2 b_1) \\|z_1 z_2| &= |z_1| |z_2| \\\arg(z_1 z_2) &= \arg(z_1) + \arg(z_2)\end{aligned}$$

What about **Complex Exponentials**?

A complex exponential is a function of type $x(t) = Ae^{zt}$ with $A = |z|$ and $z = \sigma + j\omega$

We can rewrite it as $x(t) = |z|e^{\sigma t} \cos(\omega t + \sigma) + j|z|e^{\sigma t} \sin(\omega t + \sigma)$

And thus we also have: $Ae^{j\omega t} = A \cos(\omega t) + jA \sin(\omega t)$

1.2) Signals

Signals are essentially functions on which we define some properties in order to classify them and add better and more in depth analysis tools.

Definition (Signal Classification).

- Number of independent variables
- Continuous or discrete variables

We can even identify some symmetries:

Even	Odd	Hermitian
$x(t) = x(-t)$	$x(t) = -x(-t)$	$f(x)^* = f(-x)$

any signal can be decomposed into an even and an odd part:

$$x(t) = x_e(t) + x_o(t) = \frac{x(t) + x(-t)}{2} + \frac{x(t) - x(-t)}{2}$$

We can define the following properties:

Definition (Signal Energy and Power).

We define the energy of a signal as

$$E = \int |x(t)|^2 dt$$

And the average power as

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt$$

Recall these important results:

Theorem (Notable Powers).

$$P[\cos^k(t)] = \lim_{x \rightarrow \infty} \frac{1}{2x} \int_{-x}^x \cos^{2k}(t) dt = \frac{(2k)!}{2^{2k}(k!)^2}$$

Moreover the following holds:

$$\lim_{x \rightarrow \infty} \frac{1}{2x} \int_{-x}^x \cos^k(t) dt = \begin{cases} 0, & \text{if } k \text{ odd} \\ P[\cos^{\frac{k}{2}}(t)], & \text{if } k \text{ even} \end{cases}$$

$$\lim_{x \rightarrow \infty} \frac{1}{2x} \int_{-x}^x \cos^a(\alpha t) \cos^b(\beta t) dt = 0, \text{ if one between } a, b \text{ is odd}$$

k	$\rightarrow k$	
$\cos(\alpha t)$	$\rightarrow 0$	
$A \cos^2(\alpha t)$	$\rightarrow \frac{A^2}{2}$	
$A[\cos(\alpha t) \cos(\beta t)]$	$\rightarrow 0,$	$\forall \alpha \neq \beta$
$A[\cos^2(\alpha t) \cos^2(\beta t)]$	$\rightarrow \frac{A^2}{4},$	$\forall \alpha \neq \beta$
$A[\cos^m(\alpha t) \cos^n(\beta t)]$	$\rightarrow 0,$	$\forall \alpha \neq \beta \text{ and at least one odd between } m, n$

For powers in a finite interval of one period T :

$$\begin{aligned} \cos(\alpha t) &\rightarrow 0 && \text{if } \alpha \gg 1/T \\ A \cos^2(\alpha t) &\rightarrow \frac{1}{2} A^2 T && \text{if } \alpha \gg 1/T \end{aligned}$$

The very important signals are:

Definition (Unit Sample).

This is the fundamental signal, defined by:

$$\delta(nT) = \begin{cases} \frac{1}{T} & n = 0 \\ 0 & n \neq 0 \end{cases}$$

usually we have that $T = 0$

Definition (Unit Step).

$$u[t] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Moreover $\delta[n] = u[n] - u[n - 1]$.

Recall that any discrete sequence can be represented with the unit step:

Theorem (Discrete sequence with dirac).

$$x[n] = \delta[n] * x[n] = \sum_k x[k] \delta[n - k] = \sum_k x[n - k] \delta[k]$$

1.3) Systems

A **system** is essentially an operator which can be either a known one or, what we'll actually study, more complex ones.

Some elementary systems are the ones that create a time shift/scaling and also amplitude shift/scaling.

Definition (System Classification).

- Number of input signals
- Number of output signals
- Types of processed signals

The output to a system is given by the convolution between the system and the signal

🕒 Frequency + Sampling

2) FIR Filters

In this course will mainly study two categories of filters. FIR and IIR. Let's start with the first one

Definition 1 (Finite Input Response Filters (FIR)).

FIR filters are all those filters whose **impulse response** is of finite duration

In the general case they are **causal** and with the form of a **difference equation**:

$$y[n] = \sum_{k=0}^M b_k x[n - k]$$

The **order** is defined by M and it has length $L = M + 1$.

This filter can be seen as a sliding window along $x[n]$

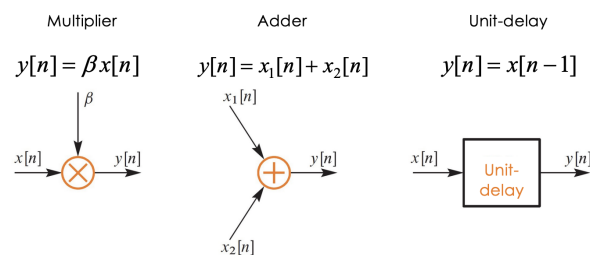
When the input to such a filter is the unit step, the result is just the coefficient of the EDO of $y[n]$ times the unit sample at that point(see [Theorem \(Discrete sequence with dirac\)](#))

Theorem 2.

A system defined by an FIR filter is LTI

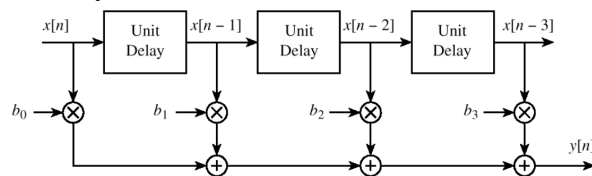
FIR Building Blocks

To implement any FIR filter we need some multipliers, adders and delayer. We can represent these using building blocks:



The most common FIR Building Blocks

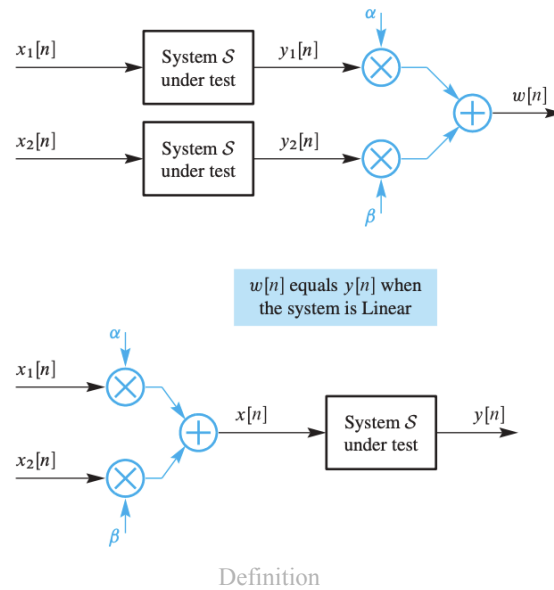
With these it is possible to represent any FIR filter



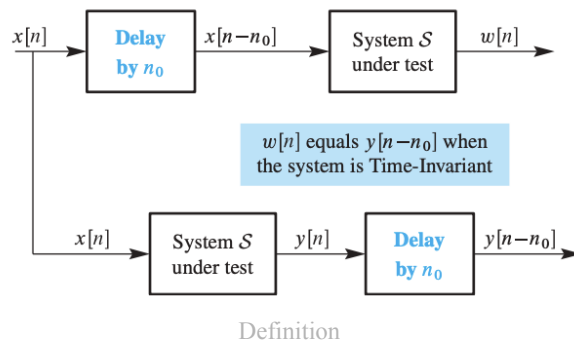
Example

With these it is also possible to give different definitions of the system properties defined in signal and systems

Definition 3 (Linearity Diagram).



Definition 4 (Time-Invariance Diagram).



2.2) LTI Systems and Convolutions

LTI stands for Linear Time Invariant. Recall that any FIR filter is LTI but not all LTI systems are FIR filters. If the input to a system is the dirac delta then the output is the filters impulse response $h[n]$: $\delta[n-l] \rightarrow h[n-l]$, moreover $x[l]\delta[n-l] \rightarrow x[l]h[n-l]$ and finally $x[n] = \sum x[l]\delta[n-l] \rightarrow y[n] = \sum x[l]h[n-l]$

Theorem 5 (LTI system properties).

- **Convolution with impulse:** $x[n] * \delta[n - n_0] = x[n - n_0]$
 - **Commutativity:** $x[n] * h[n] = h[n] * x[n]$
 - **Associativity:** $(x_1[n] * x_2[n]) * x_3[n] = x_1[n] * (x_2[n] * x_3[n])$
- These are the same properties of a convolution*

The response to a sinusoidal can be simplified when using Euler's identity to transform them into complex exponentials, so that the input is a superimposition of $x[n] = Ae^{j\phi}e^{jwn}$. With $w := wT_s$. The output will present the frequency response $H(w) = \sum b_k e^{-jwn}$ that can be also represented in polar coordinates. The following example will better show this

Example.

Let $x[n] = Ae^{j\phi}e^{jwn}$. The output to a filter with frequency response $H(w)$ will be:

$$y[n] = A|H(w)|e^{j(\phi + \angle H(w))}e^{jwn}$$

Then the output for a cosine $x[n] = A \cos(wn + \phi)$ will be:

$$y[n] = A|H(w)| \cos(wn + \phi + \angle H(w))$$

What if the input is suddenly applied with a units step?

Suppose the filter is of type M , then we can divide the signal in 3 parts: 0 if $n < 0$, the **transient** part with the first M samples and the **steady state part** with $n > M$

$$y[n] = \begin{cases} 0, & n < 0 \\ \sum_{k=0}^n (h[k]e^{-jwk})Ae^{jwn}, & 0 \leq n < M \\ \sum_{k=0}^M (h[k]e^{-jwk})Ae^{jwn}, & n \geq M \end{cases}$$

Are there any properties of the **transient component** $y_t[n]$?

- FIR: $y_t[n] = 0$ outside of bounds
- IIR: $y_t[n] \xrightarrow{n \rightarrow \infty} 0$

2.3) Frequency Response

Recall the definitions of the responses in time and frequency domain

$$h[n] = \sum_{k=0}^M h[k]\delta[n-k]$$

$$H(w) = \sum_{k=0}^M h[k]e^{-jwk}$$

Theorem 6 (Frequency Response Properties).

- *Periodicity:* $H(w + 2\pi) = H(w)$
- *Conjugate Symmetry:* if $b_k = b_k^*$, then $H(-w) = H^*(w)$. This implies that $|H(-w)| = |H(w)|$, $\angle H(-w) = \angle H(w)$ and $\Re H(-w) = \Re H(w)$, $\Im H(-w) = -\Im H(w)$ and thus the representation of $w \in [\pi, 2\pi]$ can be found by symmetry of $w \in [0, \pi]$

2.3.1) Immediate Responses

- **Delay System**

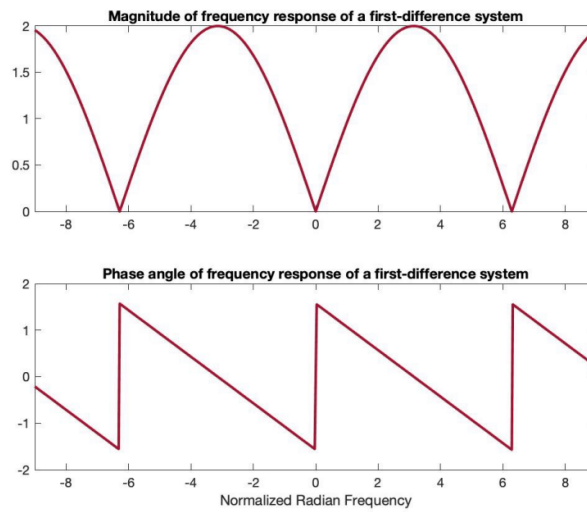
$$y[n] = x[n - n_0] \rightarrow H(w) \rightarrow e^{-jwn_0}$$

Here the magnitude is 1

- **First Difference System**

$$y[n] = x[n] - x[n - 1] \rightarrow H(w) \rightarrow 1 - e^{-jw} = 1 - \cos(w) + j \sin(w)$$

Here the magnitude is $|H(w)| = 2|\sin(w/2)|$ and the phase $\angle H(w) = \arctan(\frac{\sin(w)}{1 - \cos(w)})$



First Difference System Graphs

- **L-Point Avg Filter**

Here the computations are a bit harder, moreover we have

$$y[n] = \frac{1}{L} \sum_0^{L-1} x[n - k] \rightarrow H(w) = \frac{1}{L} \sum_0^{L-1} e^{-jwk}$$

But the sum can be rewritten as a geometric series, and thus it can be written as a **dirichlet form**

$$H(w) = D_L(w) e^{\frac{jw(L-1)}{2}} = \frac{\sin(wL/2)}{L \sin(w/2)} e^{\frac{jw(L-1)}{2}}$$

This filter has zeroes in $w = 2\pi k/L$

This analysis can be done on any type of signal

2.4) Linear Phase Filters

As the name suggests a linear phase filter is an LTI filter where the frequency response has linear phase.

The no distortion condition is given by

$$y[n] = Ax[n - n_0]$$

From here we get that the phase is in fact linear ([proof](#)). And here starts our study of linear phase filters.

Theorem 7 (Fundamental Theorem of Linear-Phase Filters).

*Since we are looking at **discrete, LTI, real and causal FIR filters** we know that the **impulse response must be symmetric** and thus*

$$h[n] = \pm h[N - n]$$

Where N is the order of the FIR filter. Only one between $\pm$ is necessary.

A lp filter satisfies the no distortion condition if N is even

[Proof](#): □

Remark.

A causal IIR system cannot satisfy these constraints (infinite and one sided thus can't be symmetric)

Types of FIR LP filters

TYPE I:

$$h[n] = h[N - n], \text{ N even}$$

Here we end up with

$$\begin{aligned}\overline{H}(\theta) &= h\left[\frac{N}{2}\right] + 2 \sum_1^{\frac{N}{2}} h\left[\frac{N}{2} - n\right] \cos(\theta n) \\ &= \sum_0^{\frac{N}{2}} p_n \cos(n\theta) \text{ with } p_0 = h\left[\frac{N}{2}\right] \text{ and } p_n = 2h\left[\frac{N}{2} - n\right]\end{aligned}$$

This is a trigonometric polynomial **symmetric** with period 2π and **symmetric** with respect to frequency $\theta = \pi$. Then $H(e^{j\theta}) = e^{-j\frac{N}{2}\theta} \overline{H}(\theta)$ with phase $\angle H(e^{j\theta}) = -\frac{N}{2}\theta + \{0, \pi\}$. This satisfies the no distortion condition.

TYPE II

$$h[n] = h[N - n], \text{ N odd}$$

Here we end up with:

$$\begin{aligned}\overline{H}(\theta) &= 2 \sum_1^{\frac{N+1}{2}} h\left[\frac{N+1}{2} - n\right] \cos\left(\left(n - \frac{1}{2}\right)\theta\right) \\ &= \sum_1^{\frac{N}{2}} p_n \cos\left(\left(n - \frac{1}{2}\right)\theta\right) \text{ with } p_n = 2h\left[\frac{N+1}{2} - n\right]\end{aligned}$$

This is a **symmetric** trigonometric polynomial with period 4π and **antisymmetric** with respect to frequency $\theta = 2\pi$.

TYPE III

$$h[n] = -h[N - n], \text{ N even}$$

Same as type I but with no term at 0 and sin instead of cos. It is **antisymmetric** and has period 2π and is **antisymmetric** to frequency $\theta = \pi$. This satisfies the no distortion condition.

TYPE IV

$$h[n] = -h[N - n], \text{ N odd}$$

As type II but with sin instead of cos. It is **antisymmetric** and has period 4π and is **symmetric** to frequency $\theta = \pi$.

2.4.2) Location of zeros for FIR LP systems

TYPE I AND II

For type I and II we have that they are symmetric $H(z) = z^{-N} H(\frac{1}{z})$ and thus the polynomial in z^{-1} is equal to its mirror image and therefore if z_i is a zero, then also $1/z_i$ is a zero.

If the zero is complex, it also includes its complex conjugate. Notice that if the zero is on the unit circle then $1/z_i = z_i^*$ and therefore there are only 2 zeroes. If it is ± 1 it is alone.

The zero at -1 is important since $H(-1) = (-1)^N H(-1)$ that is with N odd we must have a zero in -1. **Therefore Type II filters can't be used as high or band-pass filters.**

TYPE III AND IV

Here the symmetry is $H(z) = -z^{-N}H(\frac{1}{z})$ and thus $z = 1$ must be a zero. Moreover if N is even there must be a zero in -1

2.4.3) Recap

Type	Symmetries with respect to origin, $\theta = \pi$	N	No Distortion	Zeroes	Notes
I	symmetric, symmetric	Even	Yes	$z_i \implies 1/z_i$	
II	symmetric, antisymmetric	Odd	No	$z_i \implies 1/z_i, z = -1$ is a zero	Can't be high or band pass
III	antisymmetric, symmetric	Even	Yes	$z_i \implies 1/z_i, z = \pm 1$ is a zero	No low pass
IV	antisymmetric, antisymmetric	Odd	No	$z_i \implies 1/z_i, z = 1$ is a zero	No low pass

Every needed zero is a degree of freedom less from the order N of the filter. Moreover we can write the amplitude response as $\overline{H}(\theta) = Q(\theta)P(\theta)$ with $r + 1$ degrees of freedom. The values of Q and r can be found from the following table:

Type I (symmetric, N even)	$Q(\theta) = 1$	$r = \frac{N}{2}$
Type II (symmetric, N odd)	$Q(\theta) = 2\cos\left(\frac{\theta}{2}\right)$	$r = \frac{N-1}{2}$
Type III (antisymmetric, N even)	$Q(\theta) = 2\sin(\theta)$	$r = \frac{N-2}{2}$
Type IV (antisymmetric, N odd)	$Q(\theta) = 2\sin\left(\frac{\theta}{2}\right)$	$r = \frac{N-1}{2}$

P and r values

While $P(\theta) = \sum_0^r p_n \cos(n\theta)$

These filters are used in many fields, here are some examples:

- Audio Processing: equalizers, noise reduction
- Image Processing: edge detection, blurring, sharpening
- Communications: digital signal processing (DSP!) for modulation and demodulation
- Control Systems: digital systems for smoothing

Here are some final considerations:

- Pros
 - Always stable
 - Linear phase: no phase distortion since constant phase delay
- Cons
 - Much CPU intensive, IIR require less coefficients

3) Z-Transform

The Z-transform is very similar to the laplace transform, in the way that it can represent discrete time signals $x[n]$ as a complex valued function $X(z)$. There are many similarities between these 2 transforms.

Definition 8 (Z-Transform).

Let $x[n]$ be a discrete signal, then:

$$x[n] \xrightarrow{z} X(z) = \sum_{-\infty}^{\infty} x[n]z^{-n}, z \in R_x \subset \mathbb{C}$$

We can also use the unilateral z-transform where the sum starts from 0 and has a different region of convergence (ROC) R_{x+} . **if the signal is causal $x[n] = 0 \forall n < 0$ then the two transforms coincide**

Theorem 9 (ROC).

The ROC is always a ring:

$$R_x = \{z \in \mathbb{C} : R_- < |z| < R_+\}$$

Where R_- can be 0 and R_+ can be ∞ . Moreover $z = 0$ and $z \rightarrow \infty$ can also be part of the ROC. If there is no ROC then the signal doesn't have a z-transform.

- **Finite duration signal:** ROC is the entire complex plane, it excludes the origin if the last support is > 0 . It excludes points at infinity if the first support is < 0 .
- **Causal signal:** if the summation converges at $z_0 = r_0 e^{j\phi}$ then the ROC converges $\forall |z| > r_0 = R_-$. Notice that this ROC also holds for derivatives of the z-transform
- **Anticausal Signal:** Opposite of causal ROC. it converges $\forall |z| < r_0 = R_+$
- **Bilateral Signal:** this can be rewritten as sum of Causal and Anticausal signals. The ROC is the intersections of their 2 ROC.

Theorem 10 (D'Alembert determination of ROC for Causal Signals).

Consider the following inequality:

$$\frac{1}{|z|} \lim_{n \rightarrow \infty} \left| \frac{x[n+1]}{x[n]} \right| < 1$$

Then it follows that the ROC has

$$R_- = \lim_{n \rightarrow \infty} \left| \frac{x[n+1]}{x[n]} \right| < |z|$$

The same steps can be used to compute R_+ for an anticausal signal

The z-transform can be written as a ratio of 2 polynomials. These can identify zeroes and poles.

Theorem 11 (Z-Transform and Fourier Transform).

If the ROC includes the unit circle we can write the z-transform on the unit circle as

$$X(e^{j\phi}) = T \sum x[nT]e^{-jn\phi}$$

If we set $\phi = 2\pi fT = 2\pi f/F_s$ we get the Fourier Transform.

Now it is possible to understand the full connection between z and laplace transforms: Both transforms get to a form of a ratio of polynomials where zeroes and poles can be identified. Moreover the transforms can be evaluated in specific points (unit circle for z-transform and imaginary axis for laplace transform) to get to the fourier transform. The key difference is the ROC and the fact that the fourier-transform of a discrete signal has periodicity 2π which is why we consider a finite interval (unit circle) instead of an infinite interval (imaginary axis).

Properties

Linearity:	$y[n] = \alpha x[n] + \beta y[n] \xrightarrow{Z} Y(z) = \alpha X(z) + \beta Y(z),$	$R_x \cap R_y \subseteq R_w$
Time Expansion	is really complicated and thus will be omitted	
Time Delay: $n_0 > 0$	$y[n] = x[n - n_0] \xrightarrow{Z} Y(z) = z^{-n_0} X(z),$	$R_y = R_x$
Bi Time Advance: $n_0 > 0$	$y[n] = x[n + n_0] \xrightarrow{Z} Y(z) = z^{n_0} X(z),$	$R_y = R_x$
Uni Time Advance: $n_0 > 0$	$y[n] = x[n + n_0] \xrightarrow{Z} Y_+(z) = z^{n_0} X_+(z) - \sum_{n=0}^{n_0-1} x[n]z^{-n_0},$	$R_y = R_x$
Convolution:	$y[n] = x[n] * h[n] \xrightarrow{Z} Y(z) = X(z)H(z),$	$R_x \cap R_h \subseteq R_y$
Complex Conjugate:	$y[n] = \overline{x[n]} \xrightarrow{Z} Y(z) = \overline{X(\bar{z})},$	$R_y = R_x$
Time Reversal:	$y[n] = x[-n] \xrightarrow{Z} Y(z) = X(z^{-1}),$	$R_y = \{\frac{1}{R_+} < z < \frac{1}{R_-}\}$
Z Reversal:	$y[n] = (-1)^n x[n] \xrightarrow{Z} Y(z) = X(-z),$	$R_X = R_y$
Derivative:	$\frac{dX(z)}{dz} = -z^{-1} \sum nx[n]z^{-n},$	
Exponential:	$y[n] = a^n x[n] \xrightarrow{Z} Y(z) = X(\frac{z}{a}),$	$R_y = \{ a R_- < z < a R_+$
Inverse Transform:		

Theorem 12 (BIBO stability).

The z-transform is bibo stable if:

$$\sum |h[n]| < \infty, \{z \in \mathbb{C} : |z| = 1\} \subset R_h$$

that is: $h[n]$ is abs summable and has the unit circle in the roc.

Theorem 13 (Initial Value Theorem).

On the one sided z-transform we have that:

$$x[0] = \lim_{z \rightarrow +\infty} X_+(z)$$

Proof.

We know that $X_+(z) = x[0] + x[1]z^{-1} + \dots \xrightarrow{z \rightarrow \infty} x[0] + x[1] \cdot 0 \dots = x[0]$

□

Theorem 14 (Final Value Theorem).

$$\lim_{x \rightarrow \infty} x[n] = \lim_{z \rightarrow 1} (1 - z^{-1} X_+(z))$$

Proof.

Consider $y[n] = x[n] - x[n-1] \xrightarrow{\mathcal{Z}^+} Y_+(z) = X_+(z) - x[-1] - z^{-1} X_+(z)$

We also have:

$$Y_+(z) = \lim_{n \rightarrow \infty} \sum y[n] z^{-n} = \lim_{n \rightarrow \infty} \sum (x[n] - x[n-1]) z^{-n} = \lim_{z \rightarrow 1} X_+(z) (1 - z^{-1}) - x[-1]$$

Todo

□

🔍 Inverse Z

Theorem 15 (Inverse Z-transform).

Recalling the definition of Z-Transform

4) Discrete Fourier Transform (DFT)

The DFT is the Fourier transform relative to finite length discrete time-limited/periodic signals.

The periodic signal with period $T_p = NT$ is completely specified in N samples. It's Fourier transform is $X(kF)$ with $F = 1/T_p$ and periodicity $F_s = 1/T = NF$

$$X(kF) = \sum_{n=n_0}^{n_0+N-1} T x[nT] e^{-j2\pi kFnT}$$

usually we assume $n_0 = 0$.

This transform is discrete and periodic with period N

☑ **Todo**

properties

4.1) Relationship With Z-Transform and DTFT

Suppose to have a signal $x[nT]$ with support M . It will have the following z-transform:

$$X(z) = \sum_0^{M-1} x[n] z^{-n}$$

Now suppose to create $\tilde{x}[n]$ a periodic version of this signal with N samples. It's DFT is:

$$\tilde{X}(k) = \sum_0^{N-1} \tilde{x}[n] e^{-j\frac{2\pi k}{N}n}$$

Here two cases can happen.

- $N \geq M$

Clearly then the periodic signal will include all of $x[n]$ and therefore the DFT will be equal to N samples of $X(z)$ on the unit circle at equal spacing.

$$\tilde{X}(k) = \sum_0^{M-1} x[n] z_k^n \text{ with } z_k = e^{-j\frac{2\pi}{N}k}$$

These correspond to the values of the DTFT of $x[n]$ sampled at those angular frequencies. The bigger N , the more the samples

- $N < M$

Here we find temporal aliasing

4.2) Fast Fourier Transform FFT

The FFT algorithm uses the divide and conquer paradigm to reduce the computing time of the DFT to $O(N \log_2 N)$

.

The direct computation will need N complex multiplications, $N - 1$ complex additions, $N + 1$ memory accesses to read the values $x[n]$ and N accesses to read W_n (we denote W one root on the unit circle) for each coefficient, that easily equates to $O(N^2)$

With the divide and conquer we can divide the problem $\log_2 N$ times each iteration takes $O(n)$.

How to do that?

First divide the samples of $x[n]$ into odd and even samples. Then by keeping in mind that $W_N^2 = W_{\frac{N}{2}}$ we end up with two sample subsets of size $N/2$

$$X(k) = \sum_{r=0}^{\frac{N}{2}-1} x[2r] W_N^{2rk} + \sum_{r=0}^{\frac{N}{2}-1} x[2r+1] W_N^{(2r+1)k} = \sum_{r=0}^{\frac{N}{2}-1} x[2r] W_{\frac{N}{2}}^{rk} + W_N^k \sum_{r=0}^{\frac{N}{2}-1} x[2r+1] W_{\frac{N}{2}}^{rk} = G(k) + W_N^k H(k)$$

Since we are taking one sample every two for the sequences the periodicity is conserved.

The recombination step is proportional to N .

4.3) Convolution in Time Vs Frequency

In a convolution on a filter with support M each sample of $y[n]$ takes steps proportional to M . Under what circumstances is this less efficient than doing the FFT, the product and then the IFFT.

Suppose to have the signals $x[n]$, $h[n]$ and the signals given by their periodic repetition (notice that the support $N = 2^v \geq D_y$). The linear convolution will have support $D_y = D_x + M - 1$.

The frequency approach takes 2 FFT and N multiplication per sample, that is $O(\log_2 N)$

Suppose to have the times be $T_F = 3K_F \log_2 N$ and $T_D = K_D M$, then their ratio will show us if and when the frequency approach is better.

- Case $D_x \approx M$
In this case $N \approx 2M$ and the ratio shows that T_F is faster after $M = 2^4$
- Case $D_x \gg M$
Here the ratio shows that $D_x < 2^{\frac{M}{3}}$ to have the frequency approach be faster

But we can also use the **block convolution** which differs from the linear convolution since we divide the input signal in blocks. In this way we can always go back to the case where $D_x \approx M$.

4.3.1) Overlap-Add Method

Divide the signal into non overlapping blocks of length $L > M$. Assuming $x[n]$ causal and $x_k[n]$ the k -th block. we have that

$$y[n] = x[n] * h[n] = \left(\sum x_k[n] \right) * h[n] = \sum y_k[n]$$

Here the support goes from kL to $(k+1)L + M - 2$ with a total length of $L + M - 1$ and therefore there is an overlap of $M - 1$ samples. And therefore we might end up in situations where the sample is given as a sum of 2 samples. To avoid time aliasing we must have $N \geq L + M - 1$.

$$y[n] = \sum_{k=0}^{+\infty} y_k[n] = \begin{cases} y_{k-1}[n] + y_k[n] & kL \leq n \leq kL + M - 2 \\ y_k[n] & kL + M - 2 < n < (k+1)L \\ y_k[n] + y_{k+1}[n] & (k+1)L \leq n < (k+1)L + M - 2 \end{cases}$$

Sample Addition Distribution

The most optimal values are $M \approx 10^2$ and $N \approx 10^3$.

The overlap-save method is similar but there is an overlap in $x_k[n]$.

4.4) Windowing

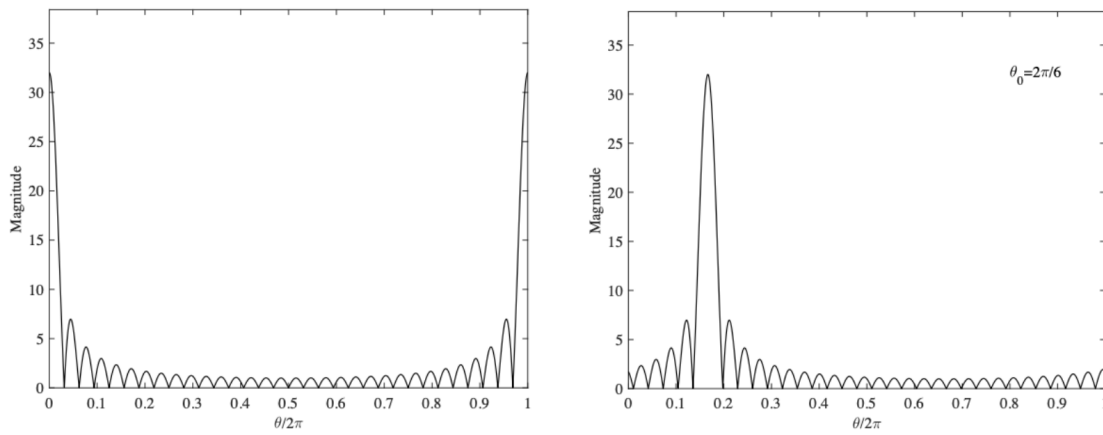
We usually cannot compute the DTFT since we can only observe a finite length section of $x[n]$.

Windowing consists in studying how the DFT of the observed signal $y[n] = x[n]w_R[n]$ is different from the DTFT of the signal.

Notice that if $x[n]$ is a complex exponential, then the DTFT is $X(e^{j\theta}) = \sum A\delta(\theta - \theta_0 + 2k\pi)$ and thus we have that

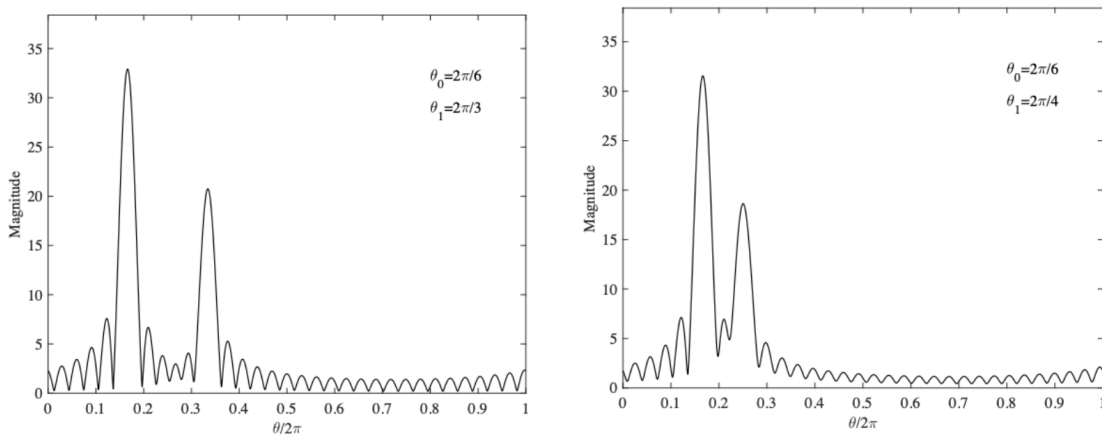
$$Y(e^{j\theta}) = AW_R(e^{j(\theta-\theta)})$$

In the case that $w[n]$ is a rect, its Fourier transform is the sinc function and thus the windowed DTFT will be a sinc centered at the frequency of the complex exponential.



Magnitude of X (left) Compared To The Magnitude of Y (right)

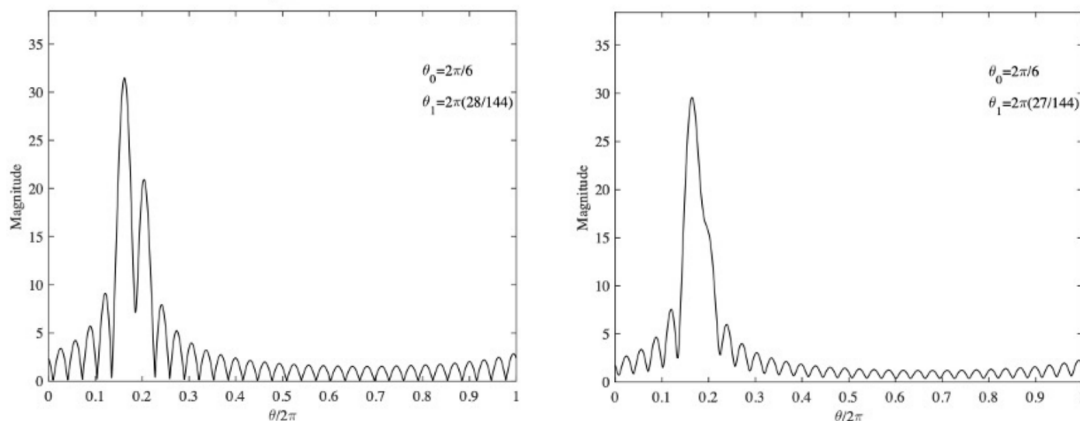
If there are more complex exponentials present in the input we will end up with 2 peaks due to superimposition. We call **leakage** the phenomenon that changes the peak of one exponential due to the contribution of the secondary peaks of the other exponentials.



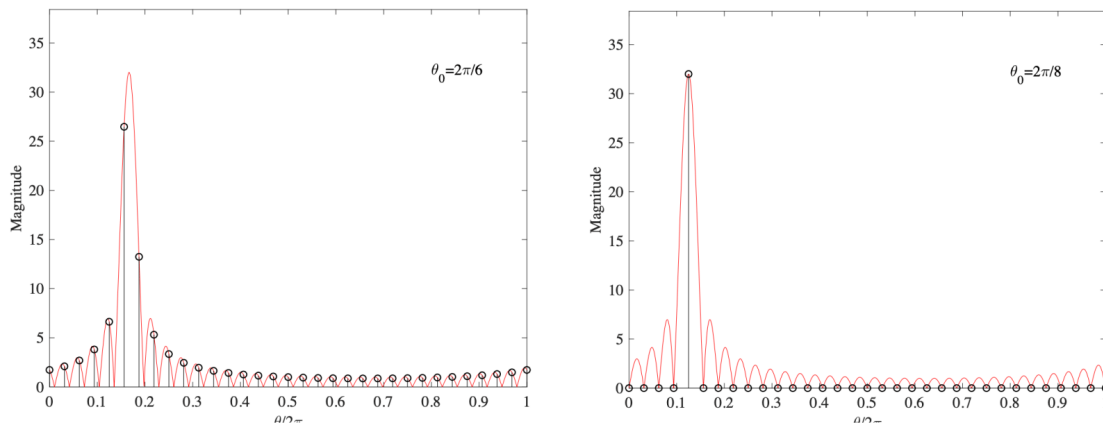
Example of Leakage

Remark 16.

If the exponentials are close in frequency they might be undistinguishable in the windowed DTFT



But as we recalled correctly the DFT is a collection of N uniformly spaced samples of the DTFT. Since the windowing shifts the we must find the integer value k_0 for which $\theta - K = \frac{2\pi}{N} k_0$ matches the peaks of the DTFT.



Wrong Vs Correct Spectral Sampling

As it is shown in the image, a correct spectral sampling can not only approximate the DTFT of the windowed signal, but even the DTFT of the non windowed signal. The accuracy can be increased by adding samples (through zero padding). Moreover it is possible to change window type in exchange for resolution to decrease leakage

Remark (Fourier Analysis of Sinusoidal Signal).

A sinusoidal signal can be expressed as a sum of complex exponentials, but we actually only need to estimate one of the two since the second one is obtained by symmetry.

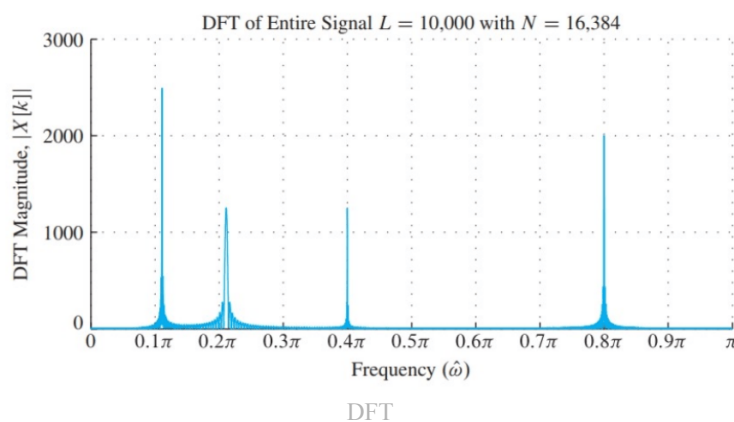
4.5) Spectrogram

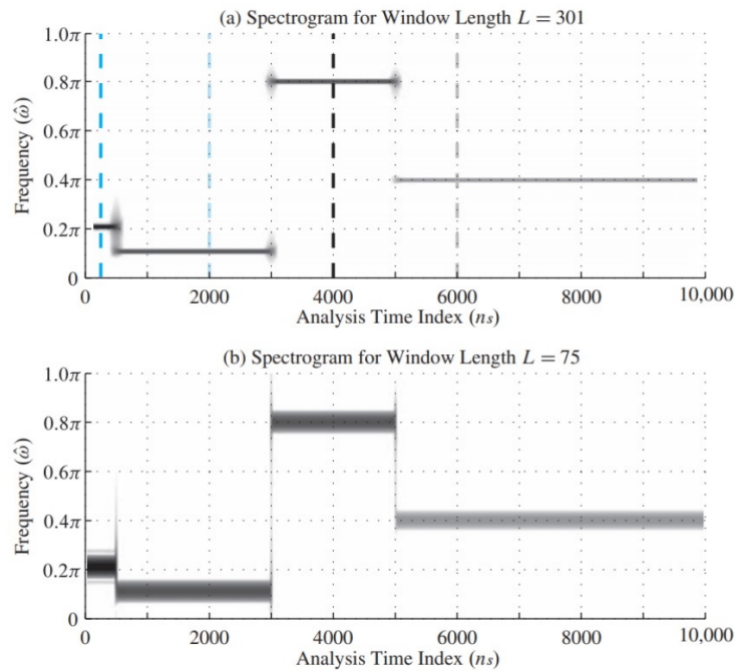
Signals that are really long ($\rightarrow \infty$) are hard to compute. The **spectrogram** is a tool that highlights the frequency difference in subsequent short sections of the signal. Therefore we need to introduce the **Short Time Discrete Fourier Transform** (STDT)

$$X[k, n_s] = \sum_{m=0}^{L-1} w[m]x[n_s + m]e^{-j(\frac{2\pi k}{N})m}$$

This encapsulates both windowing and short-length DFT. n_s is moved in jumps of $1 \leq R < L$ samples. k represents frequency and n_s represents time.

Take for example the following DFT:

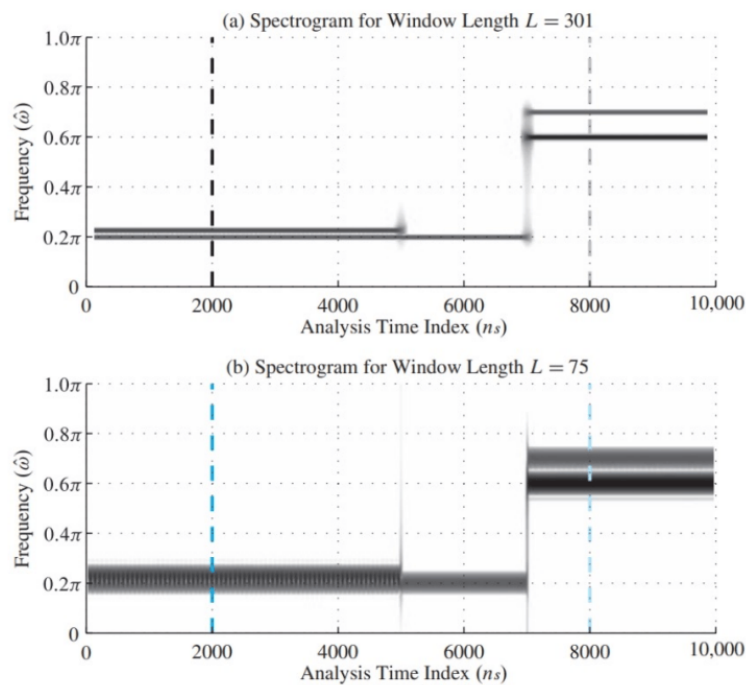




Spectrogram Representations With Different Windows

The spectrogram tells us what frequencies are present at what time. Moreover the shorter the window, the clearer it is where the frequencies change in time.

Not all signals have only one frequency at any point in time. In that case resolution is also important, since close frequencies can mix into one another.

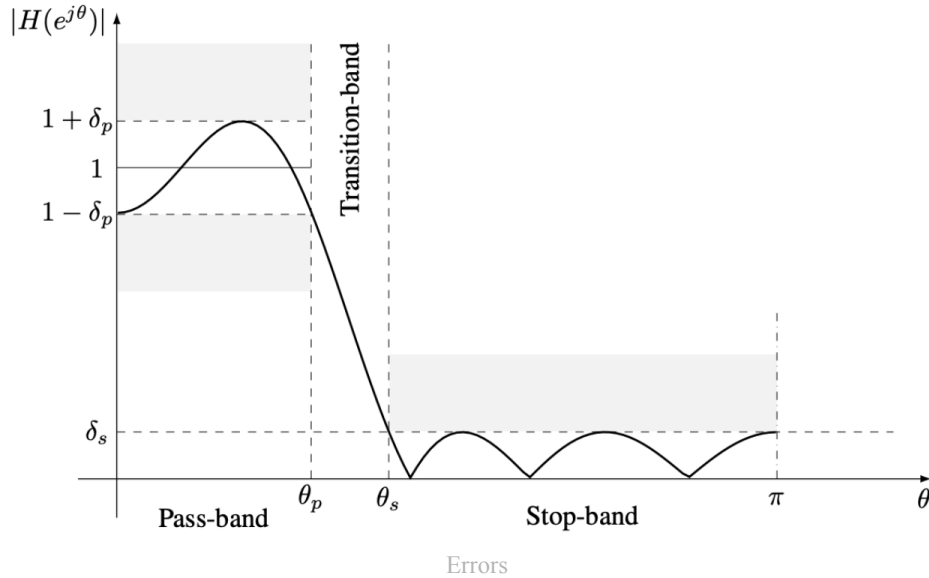


Resolution Difference

Window Size	Time Resolution	Frequency Resolution
Big	Worse	Better
Small	Better	Worse

4.6) Design Of A Low Pass Filter

A real filter won't be perfect and thus there will be some specifications on the error given.



The image represents both band errors δ and frequency limits θ .

The bands are given in dB according to the Pass Band Ripple (PBR_{dB}) and the Stop Band Attenuation (A_{dB}).

$$PBR_{dB} = 20 \log_{10} \frac{1 + \delta_p}{1 - \delta_p}$$

$$A_{dB} = -20 \log_{10} \delta_s$$

Recall that we need a linear phase filter of type I or II, that is $h[n] = \pm h[N - n]$ and the frequency response $H(\theta) = e^{j\beta} e^{-j\frac{N}{2}\theta} Q(\theta) P(\theta)$ where $\beta \in \{0, \pi/2\}$.

Let's choose type I since it has the maximum generality (no zero already set, can build any filter), and thus we can set some parameters $\beta = 0$, $Q = 1$, N even, $r = \frac{N}{2}$, $p_0 = h[\frac{N}{2}]$, $p_n = 2h[\frac{N}{2} - n]$. We strive to build this desired filter:

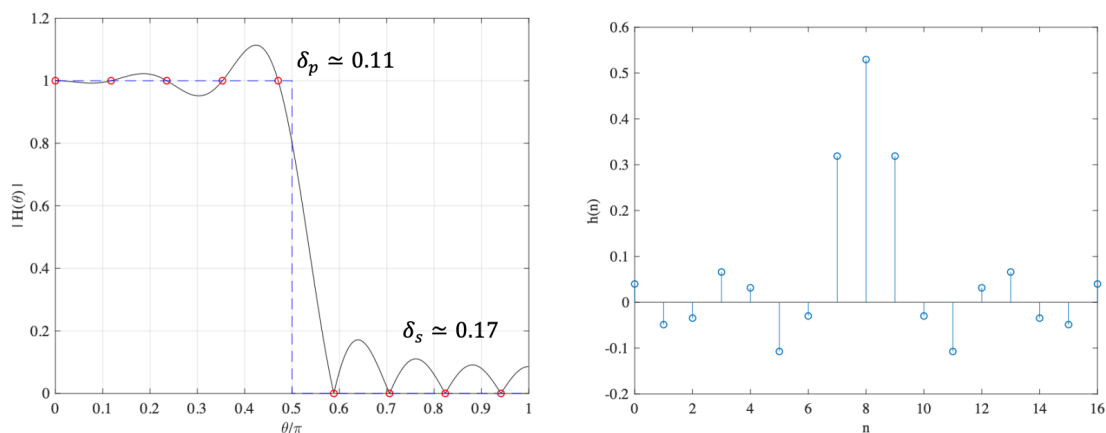
$$H_d(\theta) = e^{-j\frac{N}{2}\theta} D(\theta)$$

There are three main ways to design the filter.

4.6.1) Windowing

4.6.2) Frequency Sampling

This approach consists in filtering H_d on a set of uniformly spaced points that constitute the points of the DFT. Finally we can compute $h[n] = \text{IDFT}(H)$



Example

4.6.3) MinMaxing

We want to find the values of $h[n]$ that minimizes the maximum of the weighted error magnitude

$$E(\theta) = W(\theta)(D(\theta) - \overline{H}(\theta))$$

This is exactly as finding:

$$\min_h |E(\theta)| = \min_h \max_{\theta \in [0, \pi]} |W(\theta)(D(\theta) - \overline{H}(\theta))|$$

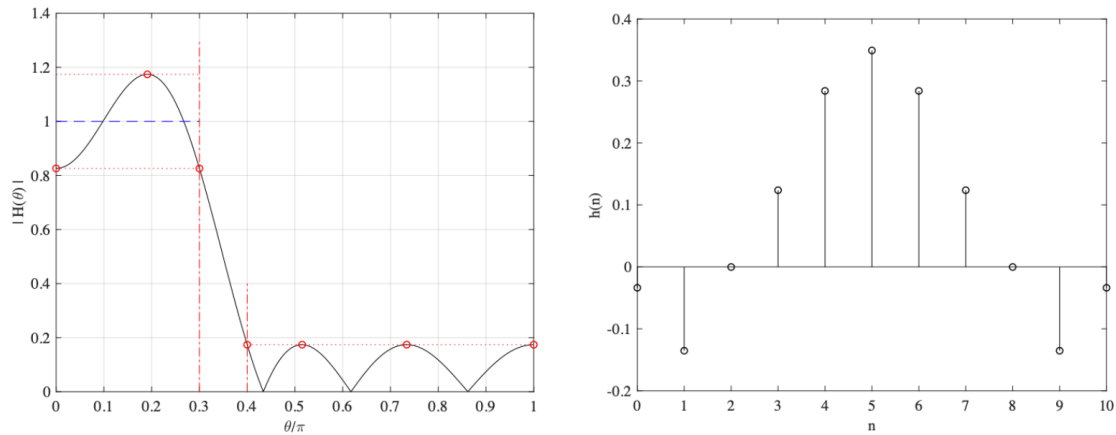
If $\overline{H}$ is a linear combination of $r + 1$ sinusoids, then E presents $r + 2$ points such that

$$|E(\theta_i)| = \max_{\theta \in [0, \pi]} |E(\theta)|$$

And these points, called **alterations**, satisfy the following equation

$$W(\theta_i) \left(D(\theta_i) - \sum_0^r p_n \cos(n\theta_i) \right) = (-1)^i \delta$$

These are essentially the $r + 2$ points that show the maximum bounds of the filter



Example

In any case the **filter order** is approximately:

$$N \approx \frac{-10 \log_{10}(\delta_p \delta_s) - 13}{14.6 \frac{\theta_s - \theta_p}{2\pi}}$$

It is **directly proportional to δ** but **doesn't depend on the cutoff frequency** and it depends on a product, thus by modifying accordingly the two values it can remain constant

5) Multirate Signal Processing

Multirate systems deal with signals that have different sampling frequencies. They are usually used to convert from one sampling frequency to another. A working assumption in this case is that the sampled signal are sampled under Nyquist conditions.

5.1) Interpolation Systems

In these systems the output sampling frequency is $L \in \mathbb{N}^+$ times bigger than the input

$$F'_s = LF_s, \quad T' = \frac{1}{L}T$$

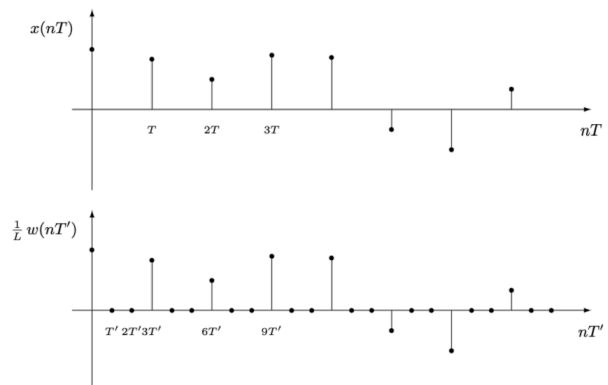
The easiest case consists of just adding $L - 1$ zero samples between 2 original samples and multiplying them by $1/L$. This system perfectly preserves all information. Moreover:

$$w[kT'] = \begin{cases} Lx[nT], & \text{if } k = nL \\ 0 & \text{otherwise} \end{cases}$$

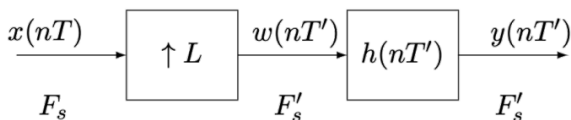
$$W(z) = X(z^L)$$

$$W(f) = X(f)$$

[Proof](#)



Example With L=3



13700.png

But this is kinda useless. Let's concatenate this with a filter to give a useful value to the new samples.

$$y[nT'] = T \sum x[kT]h[nT' - nT]$$

Where we end up having $Y(z) = H(z)X(z^L)$, $Y(f) = H(f)X(f)$.

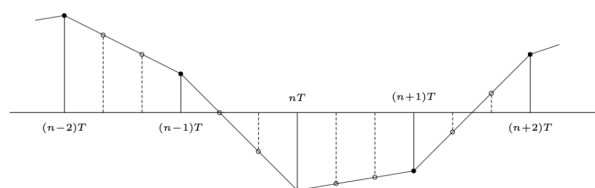
- **Ideal Low Pass Filter**

In this case $h[nT'] = F_s \text{sinc}(\frac{n}{L}) \rightarrow y[nT'] = \sum x[kT] \text{sinc}(\frac{n-kL}{L})$

This can ideally perfectly interpolate the signal, but is not physically realizable. We can approximate the filter with the windowing technique so to keep the interpolation intact.

- **Linear Interpolation**

Here the interpolation consists of tracing a line between the 2 non zero samples and projecting the zero samples onto that line.



This is a LTI system

5.2) Decimation Systems

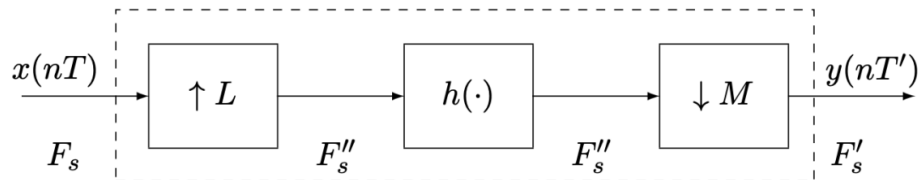
Here the output sampling frequency is $M \in \mathbb{N}^+$ times smaller

$$F'_s = \frac{1}{M} F_s, \quad T' = MT$$

5.3) Generic Rate Conversion Systems

This system is a mix of the previous two, so we can scale with rational factors

$$F'_s = \frac{L}{M} F_s, \quad T' = \frac{M}{L} T$$



Block Diagram

The system is the following: $y[nT'] = T \sum x[kT]h[nT' - kT]$ and is not time invariant

6) IIR Filters

7) Examples of Filters

7.1) Unit Delay

Theorem 17 (Unit Delay Filter).

This filter shifts the signal by an amount n_0

$$y[n] = x[n - n_0]$$

7.2) L-Point Running Average

Definition 18 (Running Average Filter).

An n-point running average filter is a filter that returns the mean of n values of the input, it can be:

- forwards:

$$y[n] = \frac{1}{l} \sum_{i=0}^l x[n + i]$$

- backwards:

$$y[n] = \frac{1}{l} \sum_{i=0}^l x[n + i]$$

- centralized:

$$y[n] = \frac{1}{l} \sum_{i=-l/2}^{l/2} x[n + i]$$

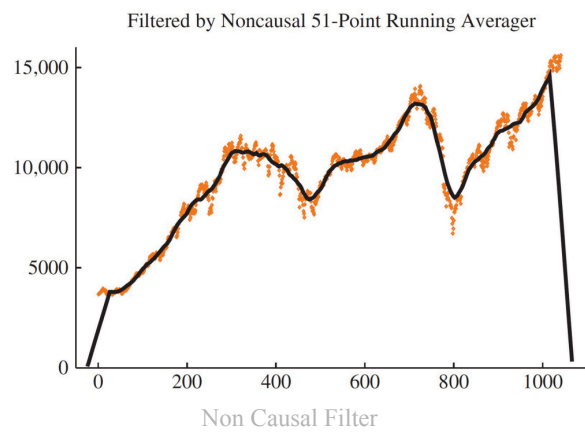
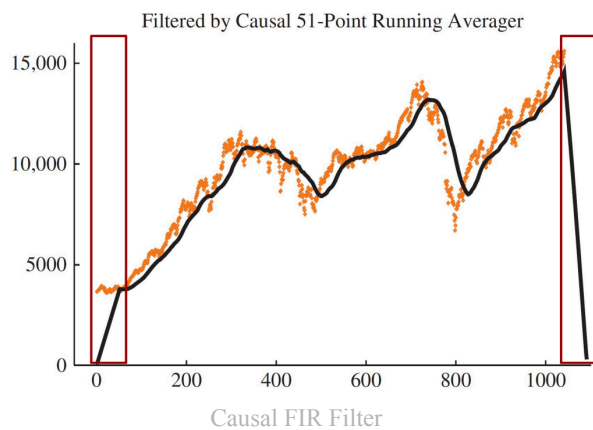
Now I will use some specific filters to highlight their properties:

Case $L = 51$:

This FIR filter is of form: $y[n] = \frac{1}{51} \sum_0^{50} x[n - k]$

This creates a shift of 25 samples as seen in the first image. We can correct this by using a non causal filter of form:

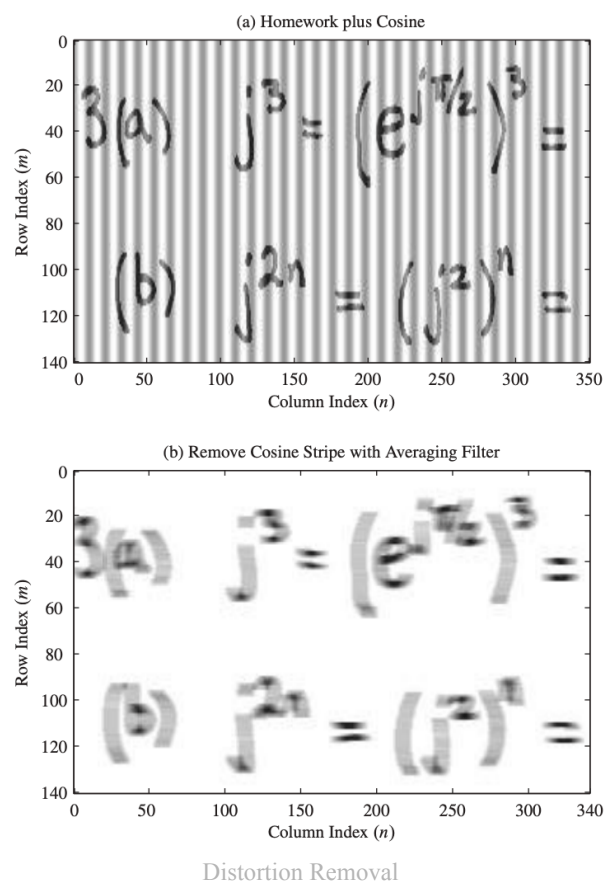
$$\tilde{y}[n] = \frac{1}{51} \sum_{-25}^{25} x[n - k]$$



Case $L = 11$:

Applying this filter on an image blurs the image in the direction it is applied. This is especially useful if the image has a cosine distortion on one axis this filter is able to remove it.

This is clearly possible if the cosine has frequency $w = 2\pi/11$ since $H(2\pi/11) = 0$ (see [L-Point Avg filter][#d39641])



7.3) Edge Detection on 2D Signals

This filter highlights edges by associating them with a high value in output magnitude. These filters are of type $y[m, n] = \sum_{k=-M}^M \sum_{l=-M}^N b_{l,k} x[m - k, n - l]$

To better see this filter consider the 1D case: $y[n] = x[n] - x[n - 1]$ here it is clear that similar values tend to cancel each other out and an increase in values comes with a positive increase

Edge-finding filters are high pass filters since edges essentially highlight high frequency changes in the image.



Example

8) Proofs

8.1) Linear Phase Filters

8.1.1) No Distortion Condition

By doing the z-transform we can find the transfer function and study phase and magnitude:

$$Y(z) = Az^{-n_0}X(z) \rightarrow H(z) = Az^{-n_0}$$

Then the phase is clearly linear $\angle H(e^{j\theta}) = -n_0\theta + \{0, \pi\}$ and thus the **group delay** is constant:

$$\tau_H = \frac{d\angle H(e^{j\theta})}{d\theta} = -n_0$$

And also the magnitude is constant $|H(e^{j\theta})| = |A|$

8.1.2) Fundamental Property of linear phase filters

Since the phase must be linear we have that $\angle H(e^{j\omega}) = \alpha\theta + \beta$ we can rewrite the frequency response using the **amplitude response** $H(e^{j\sigma}) = e^{j(\alpha\theta+\beta)}\overline{H}(\theta)$.

Since the system is real we also have that $H(e^{j\theta}) = H^*(e^{-j\theta})$. From here we see that the magnitude is even and that it is the same as $\overline{H}(\theta)$. This implies that the amplitude response is either even or odd real function. It can be expressed as $e^{j(\alpha\theta+\beta)}\overline{H}(\theta) = e^{-j(\alpha(-\theta)+\beta)}\overline{H}(-\theta)$.

- Even:

In this case $\overline{H}(\sigma) = \overline{H}(-\sigma)$ and thus we end up with $e^{j\beta} = e^{-j\beta}$. Clearly $\beta = 0$ or π . The same results are obtained for both values of β , so we will use $\beta = 0$ for the proof:

$\overline{H}(\theta) = \sum_0^N h[n]e^{-j\theta(\alpha+n)}$ but also $\overline{H}(-\theta) = \sum_0^N h[N-n]e^{-j\theta(\alpha+N+n)}$ and they are equal. This is possible with $\alpha = -N/2$.

The phase function is $\phi_H(\theta) = -\frac{N}{2}\sigma + \{0, \pi\}$

- Odd:

Here the constraints are $e^{j\beta} = -e^{-j\beta}$ then $\beta = \pm\pi/2$. With the same steps as above we have that $\alpha = -N/2$.

The phase function is $\phi_H(\theta) = -\frac{N}{2}\sigma + \frac{\pi}{2} + \{0, \pi\}$

In both cases the derivative, called **group delay** is $\alpha = -\frac{N}{2}$.

If N is even then $h[n]$ has odd symmetry with center of symmetry at one of its points. Moreover the group delay is integer and this is the delay n_0 in the no distortion condition $y[n] = Ax[n - n_0]$.

If N is odd then we don't have a self symmetric sample and the phase delay is not satisfied. The output is not only delayed but also interpolated

8.2) MSP

8.2.1) Interpolation I/O Equations

- Z-Transform:

$$W(z) = T' \sum w[nT']z^{-k} = T' \sum Lx[nT]z^{-nL} \stackrel{T=LT'}{=} T \sum x[nT](z^L)^{-n} = X(z^L)$$

Here the ROC of $W(z)$ is $R_W = \{z \in \mathbb{C} : R_-^{\frac{1}{L}} < |z| < R_+^{\frac{1}{L}}\}$

- Fourier Transform:

$$W(f) = T' \sum w(kT')e^{-j2\pi f kT'} = T' \sum Lx(nT)e^{-j2\pi f nLT'} = T \sum x(nT)e^{-j2\pi f nT} = X(f)$$

$X(f)$ has period $F_s = \frac{1}{T}$ while $W(f)$ has period $F' = \frac{1}{T'} = LF_s$